



Maths

NORTH SYDNEY BOYS HIGH SCHOOL

2007

TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write on one side of the paper (with lines) in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a new page.

- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Ireland
- ☐ Mr Barrett
- ☐ Mr Lowe
- ☐ Mr Rezcallah
- ☐ Mr Ee
- ☐ Mr Trenwith
- ☐ Mr Weiss

Student Number: /

(To be used by the exam markers only.)

Question No	1	2	3	4	5	6	7	8	9	10	Total	Total
Mark	12	12	12	12	12	12	12	12	12	12	120	100

Question 1 (12 marks)

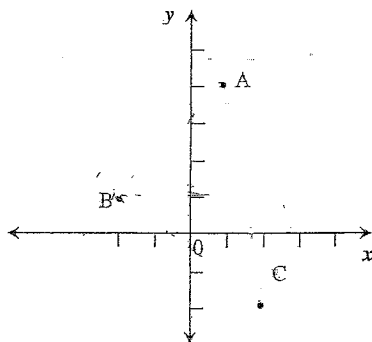
Marks

- (a) Evaluate to 2 decimal places $\frac{19 + 4.6}{\sqrt{9 - 1.4^2}}$ 2
- (b) Express $\sqrt{27} + \sqrt{48}$ in the form $a\sqrt{3}$ where a is an integer. 2
- (c) Solve for x : $|x - 4| < 6$ 2
- (d) Find the exact value of $\tan \frac{2\pi}{3}$ 2
- (e) A coin is tossed three times. Find the probability of tossing one head. 2
- (f) Solve $2x - \frac{3x + 4}{5} = 9$ 2

Question 2 (12 marks) Use a separate page

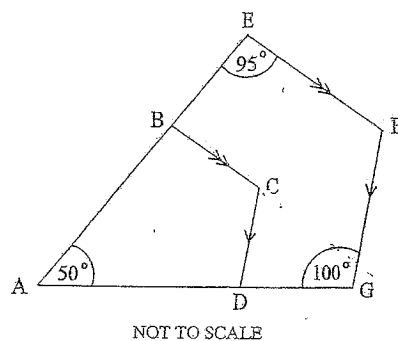
Marks

- (a) The co-ordinates of points A, B and C are A(1, 4), B(-2, 1) and C(2, -2).



- | | |
|---|---|
| (i) Show that the equation of the line AB is $x - y + 3 = 0$ | 2 |
| (ii) Find the perpendicular distance from C to AB | 2 |
| (iii) Find the distance AB. (Leave answer in simplified surd form) | 1 |
| (iv) Hence find the area of $\triangle ABC$, correct to one decimal place. | 1 |
| (v) Find the equation of the median BM in $\triangle ABC$. | 3 |

(b)



In the diagram BC is parallel to EF and CD is parallel to FG.
 $\angle BAD = 50^\circ$, $\angle BEF = 95^\circ$ and $\angle FGD = 100^\circ$.

Find the size of $\angle BCD$, giving reasons.

3

Question 3 (12 marks) Use a separate page

Marks

- (a) Differentiate with respect to x :

(i) $y = \frac{x}{e^x}$	2
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(ii) $y = (3 - \cos x)^{10}$	2
------------------------------	---

(b) (i) Evaluate $\int_0^4 e^{\frac{x}{2}} dx$	2
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(ii) Find $\int \frac{x^2}{x^3 + 2} dx$	2
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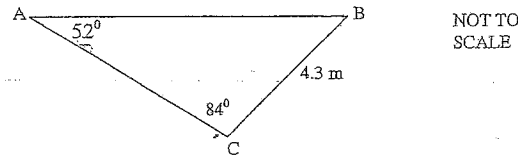
(iii) Evaluate $\int_{\frac{\pi}{3}}^{\pi} \sin 3x dx$	3
--	---

- | | |
|--|---|
| (c) Show that $\ln 2, \ln 4, \ln 8, \ln 16$ form an arithmetic sequence. | 1 |
|--|---|

Question 4 (12 marks) Use a separate page

Marks

(a)



In the diagram ABC is a triangle where $BC = 4.3$ m, $\angle ACB = 84^\circ$ and $\angle BAC = 52^\circ$.

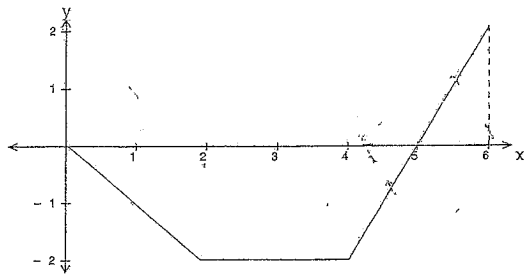
Find the length of AC correct to one decimal place.

3

(b) If $\tan^2 \theta = 3$ find the values for θ where $0 \leq \theta \leq 2\pi$

3

(c) The diagram represents a function $y = f(x)$.



Evaluate $\int_0^6 f(x) dx$.

2

(d) For the parabola $x^2 = 5y + 10x$

i) Find the focal length of this parabola.

2

ii) Calculate the coordinates of the vertex.

1

iii) Find the equation of the directrix of this parabola.

1

Question 5 (12 marks) Use a separate page

Marks

(a) Consider the curve $y = 3x^4 - 8x^3 + 6$

i) Find the turning points of the curve and determine their nature.

3

ii) Find the points of inflexion of the curve.

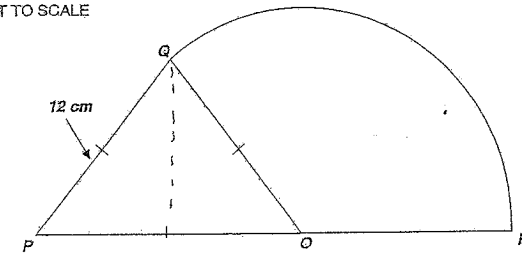
2

iii) Sketch the curve showing the turning points and the points of inflexion. (Do not find the x -intercepts).

2

(b)

NOT TO SCALE



$\triangle OPQ$ is an equilateral triangle of sides 12 cm. PR is a straight line.

QR is an arc of a circle, centre O . Find:

i) The perimeter of the whole region $PQRO$ (Leave your answer in exact form).

2

ii) The area of the whole region $PQRO$ (Leave your answer in exact form).

2

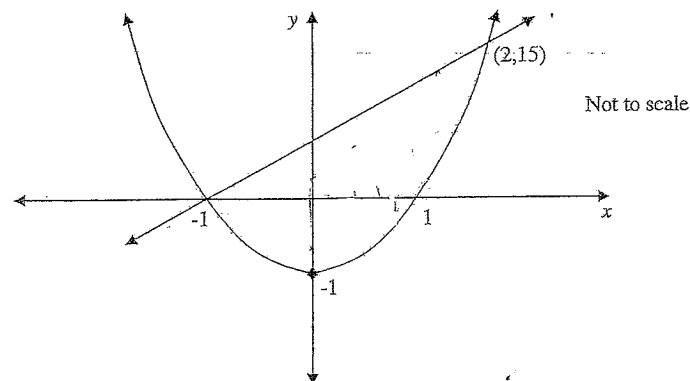
(c) Solve $\log_3(3-x) = 1$

1

Question 6 (12 marks) Use a separate page

Marks

(a)



The diagram shows the curve $y = x^2 - 1$ and the line $y = 5x + 5$

- (i) Show that the line and curve intersect at the points $(-1, 0)$ and $(2, 15)$ 2
- (ii) Calculate the area between the curve and the line. 3

- (b) A rotary club sell exactly 100 raffle tickets. When the prizes are drawn the first ticket drawn wins first prize, the second ticket drawn wins second prize and the third ticket drawn wins the third, and final, prize.

If Joseph buys two tickets, what is the probability that:

- i) he wins first prize only ? 1
- ii) he wins the first and the last prize ? 1
- iii) he wins at least one prize ? 2

- c) i) Show that the derivative of $\log_e(\tan x)$ is $\sec x \operatorname{cosec} x$. 2

- ii) Hence, evaluate $\int 2 \sec x \operatorname{cosec} x dx$ 1

Question 7 (12 marks) Use a separate page

Marks

- (a) A pool is being drained and the number of litres of water, L , in the pool at time t minutes is given by the equation

$$L = 160(60 - t)^2$$

- (i) Find the initial volume of water in the pool. 1
- (ii) At what rate is the water draining out of the pool when $t = 8$ minutes? 2
- (iii) How long will it take for the pool to be completely empty? 1

- (b) The gradient function for a curve is $\frac{dy}{dx} = 6x + \frac{1}{x}$. Find the equation of the curve if it passes through the point $(1, 8)$. 2

- (c) A plant is 65 cm high when first observed. In the first month of observation it grows 8 cm, and in each succeeding month the growth in height is 75% of the previous growth. What will be the ultimate height of the plant? 2

- (d) The region enclosed between the curve $y = \sqrt{x} + \frac{1}{\sqrt{x}}$, the x axis and the lines $x = 5$ and $x = 11$ is rotated about the x axis.

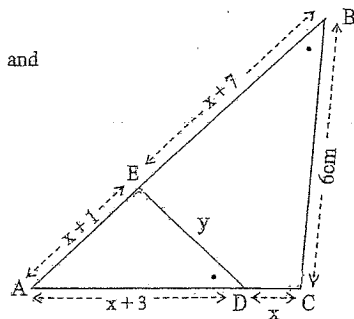
Calculate the volume of the solid generated. 4

Question 8 (12 marks) Use a separate page

Marks

- (a) In the diagram $\triangle AED$ is similar to $\triangle ACB$ and $\angle EBC = \angle EDA$.

Find the values of 'x' and 'y'.



3

- (b) Show that the equation of the tangent to $y = x \log_e x$ at the point where $x = e$ is $y = 2x - e$.

2

- (c) The roots of the quadratic equation $x^2 + (k+4)x + 5k = 0$ are α and β and $k \neq 0$.

Find the value of:

(i) $\alpha + \beta$

1

(ii) $\alpha^2 + \beta^2$

1

(iii) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$

1

- (d) A decagon (10-sided polygon) has its angle sizes in arithmetic sequence, The smallest being 117° .

(i) Show that the angle sum of the decagon is 1440° .

1

(ii) Find the common difference of the sequence.

2

(iii) Find the size of the largest angle.

1

Question 9 (12 marks) Use a separate page

Marks

- (a) Find the values of k for which the quadratic equation $x^2 - (k+2)x - 3k - 11 = 0$ has real roots.

3

- (b) The value \$V of a machine is given by the formula $V = Ae^{-kt}$, where A and k are constants and t is the time measured in years. The initial cost of the machine is \$120 000. Two years later the value was \$95 000.

(i) Calculate the value of the constants A and k .

2

(ii) Calculate the value of the machine after 10 years.

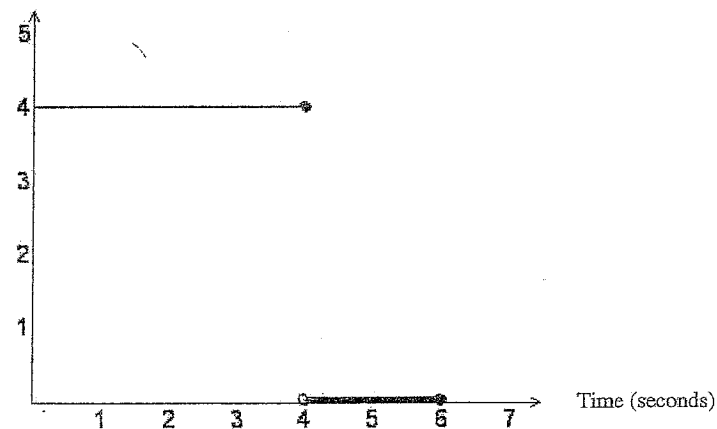
1

(iii) If the machine was bought on 1st January 2000, in which year will the value of the machine fall below \$60 000?

2

- (c) This graph shows the acceleration of a particle during a 6 second interval.

Acceleration m/s^2



Initially the particle is at rest at the origin.

- i) Sketch, accurately, the velocity time graph of the particle for $0 \leq t \leq 6$.

2

- ii) Calculate the distance travelled by the particle during this 6 seconds.

2

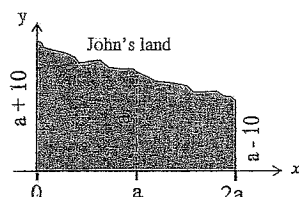
Question 10 (12 marks) Use a separate page

Marks

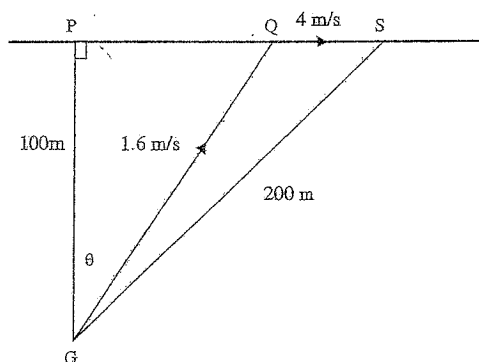
STANDARD INTEGRALS

- a) The shaded area shown in the diagram below represents John's land. Its dimensions are given in terms of 'a'.

Given that the area of this land is 3200m^2 , use Simpson's rule with 3 function values to find an estimate the value of 'a'.



- b) A swimmer at G in a calm lagoon is 100 metres from the beach at P when he decides to return for a snack to S, which is on the beach 200 metres from him. He can swim at 1.6 metres per second and jog along the beach at 4 metres per second.



$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x, x > 0$

- i) Show that $QS = 100(\sqrt{3} - \tan \theta)$ where $\theta = \angle PGQ$ 2

- ii) Show that the total time of travel T (in seconds) for the trip sections is given by: 2

$$T = \frac{125}{2 \cos \theta} + 25(\sqrt{3} - \tan \theta)$$

- iii) In which direction θ should he start swimming to reach the snack in the shortest time? 4

- iv) How long till he reaches his destination? 1



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EXAMINATION

Mathematics

Solutions - Marking scale

Marking:

Q1 & Q2 – Mr Ireland

Q3 & Q4 – Mr Ireland

Q5 & Q6 – Mr Barrett

Q7 & Q8 – Mr Lowe

Q9 & Q10 – Mr Rezcallah

Marking Guidelines: Mathematics Solutions

Answers:

Question 1

$$\frac{19 + 4.6}{\sqrt{9 - 1.4^2}} = 8.89458$$

1.(a) $= 8.89$ (2 dec places)

(b) $\sqrt{27} + \sqrt{48} = 3\sqrt{3} + 4\sqrt{3} = 7\sqrt{3}$

(c) $|x - 4| < 6$
 $\therefore -6 < x - 4 < 6$
 $-2 < x < 10$

(d) $\tan \frac{2\pi}{3} = \tan \left(\pi - \frac{\pi}{3} \right) = -\tan \frac{\pi}{3} = -\sqrt{3}$

(e) outcomes possible are HTT, THT, TTH
 since there are 8 outcomes $\therefore P(1H) = \frac{3}{8}$

(f) $2x - \frac{3x+4}{5} = 9$
 $10x - 3x - 4 = 45$
 $7x = 49$
 $x = 7$

Question 2

2(a)

(i) $y - 4 = \frac{1-4}{-2-1}(x-1)$
 $y - 4 = x - 1$ where $m = 1$
 $0 = x - y + 3$

(ii) $d = \frac{|1 \times 2 - 1 \times -2 + 3|}{\sqrt{1^2 + (-1)^2}}$
 $= \frac{|7|}{\sqrt{2}} = \frac{7}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{7\sqrt{2}}{2}$

Question 1

(a)
 1 for approximation
 1 f or 2 dp

(b)
 1 for simplifying a surd
 1 for the simplification.

(c)
 1 for end points
 1 for correct inequality

(d)
 1 for $-\tan \frac{\pi}{3}$
 1 for exact answer

(e)
 1 for indicating 8 outcomes
 1 for correct answer

(f)
 1 for multiplying by 5 and correct signs

1 for answer $x = 7$

Question 2

(a)(i) 1 for gradient $m=1$

1 for correct equation formula to get the given answer

(ii) 1 for sub. into formula

1 for correct answer

(iii) $d = \sqrt{(-2-1)^2 + (1-4)^2}$ $= \sqrt{18}$ $= 3\sqrt{2}$	(iii) 1 for correct answer (do not penalize for decimal answer if formula shown)
(iv) $A = \frac{1}{2} \times 3\sqrt{2} \times \frac{7\sqrt{2}}{2}$ $= 10.5\text{u}^2$	(iv) 1 for correct answer
(v) Mid point M of AC: $M(\frac{1+2}{2}, \frac{4-2}{2})$ $= M(\frac{3}{2}, 1)$ Gradient of BM: $m = \frac{1-1}{\frac{3}{2}+2} = 0$	(v) 1 for correct midpt M 1 for $m = 0$
Equation of BM : $y = 1$	1 for equation $y = 1$ Award full marks if students show with valid reasoning, drawings or workings that $y = 1$ N.B: 2 marks only if another median is found correctly
(b) $\angle BAD = 50^\circ$ $\angle ABC = \angle BEF = 95^\circ$ (corresponding $\angle$ s of // $\angle ADC = \angle FGD = 100^\circ$ lines are $\Rightarrow$) $\angle BCD = 360^\circ - (50^\circ + 95^\circ + 100^\circ)$ ($\angle$ sum of quad $= 115^\circ$ ABCD)	(b) 1 for correct values of $\angle BEF$ and $\angle FGD$ 1 for correct reason(s) 1 for correct $\angle BCD$ value with working or reason

Answers:

Question 3	3 (a)
3(a)	(i)
(i) $y = \frac{x}{e^x}$ $y' = \frac{1(e^x) - x(e^x)}{(e^x)^2}$ $= \frac{(1-x)(e^x)}{e^x \cdot e^x} = \frac{(1-x)}{e^x}$ Accept also product rule answer: $y = e^{-x} - xe^{-x}$	1 for applying correct rule 1 for correct answer
(ii) $y = (3 - \cos x)^{10}$ $y' = 10(-\sin x)(3 - \cos x)^9$ $= 10(\sin x)(3 - \cos x)^9$	(ii) 1 for applying chain rule 1 for simplification
(b)	(b)
(i) $\int_0^4 e^{\frac{x}{4}} dx = \left[4e^{\frac{x}{4}} \right]_0^4 = 4e - 4$	(i) 1 for integrating 1 for correct answer
(ii) $\int \frac{x^2}{x^3+2} dx = \frac{1}{3} \int \frac{3x^2}{x^3+2} dx$ $= \frac{1}{3} \log_e(x^3+2) + c$	(ii) 1 for correct method 1 for correct answer
(iii) $\int_{\frac{\pi}{3}}^{\pi} \sin 3x dx = \left[\frac{-\cos 3x}{3} \right]_{\frac{\pi}{3}}^{\pi}$ $= \left(\frac{-\cos 3\pi}{3} \right) - \left(\frac{-\cos \pi}{3} \right)$ $= \frac{1}{3} - \frac{1}{3}$ $= 0$	(iii) 1 for integrating 1 for substituting 1 for correct answer
(c) $t_2 - t_1 = t_3 - t_2 \Rightarrow \log 4 - \log 2 = \log 8 - \log 4$ $2\log 2 - \log 2 = 3\log 2 - 2\log 2$ $\log 2 = \log 2$ $\therefore$ an arithmetic sequence	(c) 1 for correctly showing it's AS

Answers

Question 4

4 (a)

(i) $\angle ABC = 44^\circ$

$$\frac{AC}{\sin 44^\circ} = \frac{4.3}{\sin 52^\circ}$$

$$AC = \frac{4.3 \sin 44^\circ}{\sin 52^\circ}$$

$$= 3.790\dots$$

$$= 3.8 \text{ (1dp)}$$

(b) $\tan^2 \theta = 3 \therefore \tan \theta = \pm \sqrt{3}$

$$\tan \theta = \sqrt{3} \Rightarrow \theta = \frac{\pi}{3}, \pi + \frac{\pi}{3}$$

$$= \frac{\pi}{3}, \frac{4\pi}{3}$$

$$\tan \theta = -\sqrt{3} \Rightarrow \theta = \pi - \frac{\pi}{3}, 2\pi - \frac{\pi}{3}$$

$$= \frac{2\pi}{3}, \frac{5\pi}{3}$$

(c)

For $\int_0^6 f(x) dx$: students can identify that the solution simplifies to be the area of a trapezium under the x-axis.

$$\text{Hence } \int_0^6 f(x) dx = -\left\{\frac{2}{2}(4+2)\right\}$$

$$= -6$$

(d)

$$x^2 = 5y + 10x$$

$$x^2 - 10x + 25 = 5y + 25$$

$$(x-5)^2 = 5(y+5) \text{ of the form}$$

$$(x-h)^2 = 4a(y-k)$$

$$i) 4a = 5 \Rightarrow \text{focal length} = \frac{5}{4}$$

$$ii) h = 5, k = -5$$

$$\text{Vertex } (5, -1)$$

$$iii) \text{Directrix } y = -6 \frac{1}{4} = -\frac{25}{4}$$

Question 4

4 (a)

1 using correct $\angle ABC$

1 using sine rule correctly

1 for correct answer

(b)

1 for finding 2 equations

1 for correct 2 answers

1 for other 2 correct answers

(c)

1 for finding areas between the curve and the x-axis (above & below); or simplifies the question as in solution

1 for correct final answer of -6.

(d)

1 for changing the equation to the known form

i) 1 for correct a

ii) 1 for correct V

iii) 1 for correct directrix

Answers:

Question 5

5 (a)

$$i) y = 3x^4 - 8x^3 + 6$$

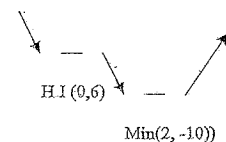
$$\frac{dy}{dx} = 12x^3 - 24x^2 = 0$$

$$12x^2(x-2) = 0$$

$$\therefore x = 0, 2$$

x	-1	0	1	2	3
$\frac{dy}{dx}$	-36	0	-12	0	108

y



The curve has a horizontal point of inflexion at (0,6) and a Minimum at (2,-10);

ii)

$$\frac{d^2y}{dx^2} = 36x^2 - 48x = 0$$

$$12x(3x-4) = 0$$

$$\therefore x = 0, \frac{4}{3}$$

x	-1	0	1	4/3	2
y''	84	0	-12	0	48

y

Conc up Conc down Conc up

The curve has a horizontal point of inflexion at (0,6)

and another point of inflexion at $(\frac{4}{3}, -\frac{13}{27})$

Question 5

a)

i)

1 for correct derivative = 0

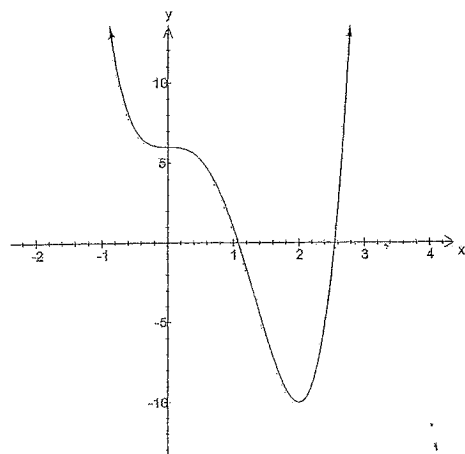
1 for testing each of the 2 points

1 for each point's coordinates and nature

1 for finding each of the 2 inflexions

1 for testing the non zero inflexion

iii)



The 3 points should be labeled on the diagram.

b) i)

$$\angle QOR = 120^\circ \text{ (straight line)}$$

$$\text{Arc } QR = r\theta$$

$$= 12 \left(\frac{2\pi}{3} \right) \\ = 8\pi$$

$$\therefore \text{Perimeter} = 12 + 12 + 12 + \text{arc } QR \\ = 36 + 8\pi \text{ cm}$$

ii)

$$\text{Area} = \text{Triangle} + \text{Sector}$$

$$= \frac{1}{2} ab \sin C + \frac{1}{2} r^2 \theta$$

$$= \frac{1}{2} \cdot 12 \cdot 12 \cdot \sin 60^\circ + \frac{1}{2} \cdot 12^2 \cdot \frac{2\pi}{3}$$

$$= \frac{1}{2} \cdot 12 \cdot 12 \cdot \frac{\sqrt{3}}{2} + \frac{1}{2} \cdot 144 \cdot \frac{2\pi}{3}$$

$$= 36\sqrt{3} + 48\pi \text{ cm}^2$$

iii)

$$\log_3(3-x) = 1 = \log_3 3$$

$$3-x=3$$

$$x=0$$

iii) for correct shape

1 for the 3 points found earlier shown on the diagram

b) i)

1 for correct arc QR

1 for correct answer or their length of arc QR + 36.

ii)

1 for correct answer for area of triangle.

1 for correct answer for area of sector (using their calculation for $\angle QOR$) in radians.

iii)

1 for correct answer

Answers

Question 6

6 (a)

(i) sub $(-1, 0)$ and $(2, 15)$ into $y = x^4 - 1$

$$0 = 1 - 1 \text{ true and } 15 = 16 - 1 \text{ true}$$

sub $(-1, 0)$ and $(2, 15)$ into $y = 5x + 5$

$$0 = -5 + 5 \text{ true and } 15 = 10 + 5 \text{ true}$$

Or use algebra (factorization)

$$(ii) A = \int_{-1}^2 (5x + 5) - (x^4 - 1) dx$$

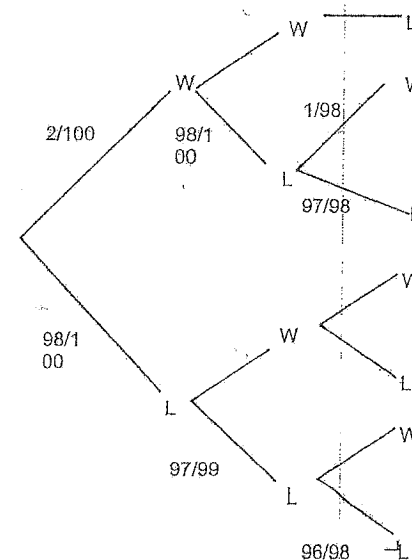
$$= \int_{-1}^2 -x^4 + 5x + 6 dx$$

$$= \left[\frac{-x^5}{5} + \frac{5x^2}{2} + 6x \right]_{-1}^2$$

$$= \left(\frac{-32}{5} + 10 + 12 \right) - \left(\frac{1}{5} + \frac{5}{2} - 6 \right)$$

$$= 15 \frac{3}{5} - \left(-3 \frac{3}{10} \right) = 18.9 u^2$$

(b)



Question 6

(6) (a)

(i)

1 for showing $(-1, 0)$ lies on the intersection.

1 for showing $(2, 15)$ lies on the intersection.

(ii)

1 correct integration

1 for correct sub.

1 for correct answer

Question 6(b) Continued

(i)

$$P(WLL) = \frac{2}{100} \times \frac{98}{99} \times \frac{97}{98} = \frac{97}{4950}$$

(ii)

$$P(WLW) = \frac{2}{100} \times \frac{98}{99} \times \frac{1}{98} = \frac{1}{4950}$$

(iii)

$$P(\text{win at least 1 prize}) = 1 - P(\text{no prizes})$$

$$= 1 - P(LLL) = 1 - \frac{98}{100} \times \frac{97}{99} \times \frac{96}{98} = \frac{49}{825}$$

(c) i) $y = \log_e(\tan x)$

$$\frac{dy}{dx} = \frac{\sec^2 x}{\tan x}$$

$$= \frac{1 + \tan^2 x}{\tan x} = \frac{1}{\tan x} + \tan x$$

$$= \frac{\cos x}{\sin x} + \frac{\sin x}{\cos x}$$

$$= \frac{\cos^2 x + \sin^2 x}{\sin x \cos x} = \frac{1}{\sin x \cos x}$$

$$= \sec x \csc x$$

Alternatively,

$$\frac{dy}{dx} = \frac{\sec^2 x}{\tan x}$$

$$= \frac{\sec^2 x}{\tan x} = \frac{1}{\cos^2 x} \div \frac{\sin x}{\cos x}$$

$$= \frac{1}{\cos^2 x} \times \frac{\cos x}{\sin x}$$

$$= \frac{1}{\sin x \cos x}$$

$$= \sec x \csc x$$

ii) $\int 2 \sec x \csc x dx =$
 $2 \ln(\tan x) + C$

Question 6(b)

(i)
 1 for correct answer

(ii)
 1 for correct answer
 (iii)
 1 for correct method

1 for correct answer

(c) i)
 1 for correct derivative

1 for correct method

ii) 1 for correct integration

Question 7

7(a)

(i) $L = 160(60 - t)^2$

let $t = 0$

$\therefore L = 160(60 - 0)^2$

$\therefore L = 576000 \text{ Litres}$

(ii) $\frac{dL}{dt} = -320(60 - t)$

when $t = 8$

$\frac{dL}{dt} = -16640$

$\therefore$ draining out at 16640 litres per minute

(iii) $0 = 160(60 - t)^2$

$\therefore t = 60 \text{ minutes}$

(b)

$\frac{dy}{dx} = 6x + \frac{1}{x}$

$y = \int (6x + \frac{1}{x}) dx = 3x^2 + \ln x + c$

Sub in (1,8)

$8 = 3(1) + 0 + c$

$c = 5$

$y = 3x^2 + \ln x + 5$

(c) Application of Limiting Sum

height = $63 + 3 + \frac{3}{4} \times 3 + \frac{3}{4} \times \frac{3}{4} \times 3 + \dots$

$= 63 + \frac{3}{1 - \frac{3}{4}}$

$= 63 + 32 = 95 \text{ cm}$

Question 7

(7) (a)

(i)

1 for correct answer

(ii)

1 for correct derivative

1 for correct answer

(iii)

1 for correct answer

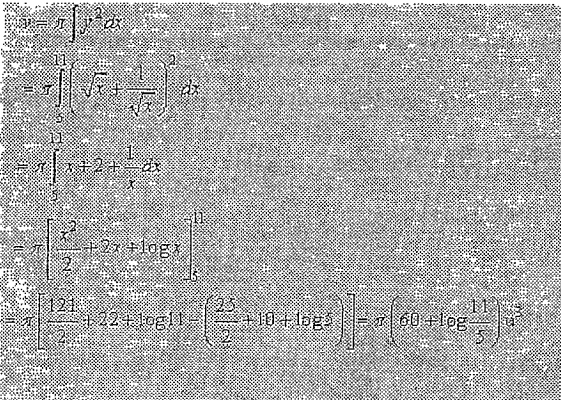
1 for correct integration

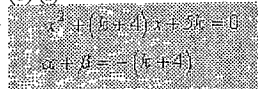
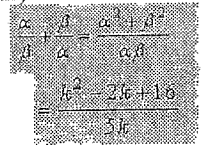
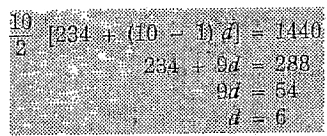
1 for correct c value

(c)

1 for establishing the G.P.

1 for correct answer

Question 7 Continued (d)  Accept also $V=190.97 \text{ u}^3$	Question 7 Continued (d) 1 for correct integration statement 1 for correct integration 1 for substitution 1 for simplification.
Question 8 a) Since $\triangle AED$ is similar to $\triangle ACB$ $\frac{AE}{AC} = \frac{AD}{AB}$ $\frac{x+1}{2x+3} = \frac{x+3}{2x+8}$ $(x+1)(2x+8) = (x+3)(2x+3)$ $2x^2 + 10x + 8 = 2x^2 + 9x + 9$ $\therefore x = 1 \text{ cm}$ $\frac{ED}{BC} = \frac{AD}{AB}$ $\frac{y}{6} = \frac{x+3}{2x+8} = \frac{4}{10}$ $10y = 24$ $y = 2.4 \text{ cm}$	1 for correct ratio in terms of x 1 for correct answer $x=1 \text{ cm}$ 1 for correct answer $y=2.4 \text{ cm}$

(b) $y = x \ln x$ $y' = \ln x + x \left(\frac{1}{x} \right) = \ln x + 1$ $m = \ln e + 1 = 1 + 1 = 2$ At $x = e$, $y = e \ln e = e$ $y - e = 2(x - e)$ $y = 2x - 2e + e = 2x - e$ (c) (i)  ii)  $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$ $= (k+4)^2 - 10k$ $= k^2 - 2k + 16$ (iii)  (d)(i) Angle sum = $(10 - 2) 180^\circ$ $= 1440^\circ$ (ii)  (iii) $T_{10} = 117 + (10 - 1) 6$ $= 171$	(b) 1 for correct derivative 1 for showing the given equation (c)(i) 1 mark for correct answer (ii) 1 mark for correct answer (iii) 1 mark for correct answer (d)(i) 1 for showing the angle sum (ii) 1 for correct application of the sum formula 1 mark for correct d (iii) 1 mark for correct answer
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Answers

Question 9

(9)(a)

$$(k+2)x^2 - 2kx - 11 = 0$$

for real roots $\Delta \geq 0$

$$(k+2)^2 - 4 \times 1 \times (-11) \geq 0$$

$$k^2 + 4k + 4 + 44 \geq 0$$

$$k^2 + 4k + 48 \geq 0$$

$$(k+12)(k+4) \geq 0$$

$$k \leq -12, k \geq -4$$

(b)

(i) $V = Ae^{-kt}$

$t = 0, V = 120\,000 \therefore A = 120\,000$

$t = 2, V = 95\,000$

$\therefore 95\,000 = 120\,000e^{-2k}$

$\frac{19}{24} = e^{-2k}$

$\log \frac{19}{24} = -2k$

$\therefore k = -\frac{1}{2} \log \frac{19}{24}$

$= 0.1168..$

$= 0.117(3dp)$

(ii) When $t = 10$

$V = 120\,000e^{-10 \times 0.1168..}$

$= 37\,315.82$

(iii)

$120\,000e^{-kt} < 60\,000$

$e^{-kt} < \frac{1}{2}$

$-kt < \log \frac{1}{2}$

$\log \frac{1}{2}$

$t > \frac{\log \frac{1}{2}}{-k}$

$t > 5.92$

$\therefore$ during 2005

Question 9

(a) 1 for $\Delta \geq 0$

1 for quad. inequation

1 for correct inequality answers.

(b)

(i) 1 for A

1 for k

(ii) 1 for correct ans.

(iii)

1 for taking logs

1 for answer

(c) i)

For $0 < t < 4$, $a = 4 \text{ m/s}^2$

$v = 4t + c$

@ $t = 0$, $v = 0 \therefore c = 0$

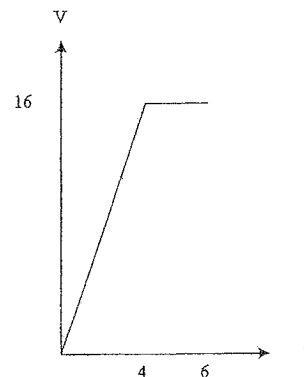
$\therefore v = 4t$ for $0 < t < 4$

@ $t = 4$, $v = 4(4) = 16 \text{ m/s}$

Alternatively to find the velocity @ $t = 4$ s,

$v =$ Area of rectangle of sides 4 by 4 = 16 m/s. Then the particle travels with this constant velocity as $a = 0$ when $4 < t < 6$.

$\therefore v = 16 \text{ m/s}$ for $4 < t < 6$



ii)

$x = A_1 + A_2$

$= \frac{4 \times 16}{2} + 16 \times 2$

$= 32 + 32 = 64 \text{ m}$

Alternatively,

$x = \int_0^4 4t dt + \int_4^6 16 dt$

$= \left[2t^2 \right]_0^4 + \left[\frac{16t}{1} \right]_4^6$

$= 32 + 16(6) - 16(4)$

$= 32 + 32$

$= 64 \text{ m}$

(c) i)

1 for $v = 16 \text{ m/s}$

1 for correct graph with 0 and 16 shown clearly in the 2 phases of motion

1 for Area of one section correct

1 for correct answer

Answers

Question 10

a) $h=a$

x	0	a	$2a$
y	$a+10$	a	$a-10$

$$A = \frac{h}{3}(y_1 + 4y_2 + y_3)$$

$$3200 = \frac{a}{3}(a+10+4a+a-10)$$

$$3200 = \frac{a}{3}(6a)$$

$$2a^2 = 3200$$

$$a^2 = 1600$$

$$a = \pm 40 \quad (a > 0)$$

$$\Rightarrow a = 40$$

Question 10

1 for correct
simpson's
equation
applied in this
question

1 for solving
equation

1 for $a = 40$
(0 for $\pm$)

b) (i)
1 for PS

1 for PQ

(ii)

1 for GQ

1 for showing
the given time
equation

b) (i)

Also, $PQ + QS = PS$ where $PS^2 = 200^2 - 100^2 = 30000$ (Pythagoras theorem)

Hence $PS = 100\sqrt{3}$ and as $PQ = 100 \tan \theta$ we find $QS = 100(\sqrt{3} - \tan \theta)$.

(ii)

$$\text{Now } 100 = GQ \cos \theta \text{ so } GQ = \frac{100}{\cos \theta}$$

The time of travel T (in seconds) is given by $\frac{\text{distance travelled}}{\text{speed}}$ for the trip sections

$$T = \frac{GQ}{1.6} + \frac{QS}{4}$$

$$T = \frac{100}{1.6 \cos \theta} + \frac{100}{4}(\sqrt{3} - \tan \theta)$$

$$T = \frac{125}{2 \cos \theta} + 25(\sqrt{3} - \tan \theta)$$

(iii)

$$\frac{dT}{d\theta} = \frac{100}{1.6} \left(\frac{\sin \theta}{\cos^2 \theta} \right) - \frac{25}{\cos^2 \theta}$$

for a stationary point

$$\Rightarrow 100 \sin \theta = 25 \times 1.6$$

$$\Rightarrow \sin \theta = 0.4$$

$$\text{so } \theta = 23.57^\circ \text{ or } = 23^\circ 35'$$

Testing:

$$\theta = 0, \quad \frac{dT}{d\theta} = 0 - \frac{25}{1} = -25$$

$$\theta = 30, \quad \frac{dT}{d\theta} = \frac{100}{1.6} \left(\frac{0.5}{0.75} \right) - \frac{25}{0.75} \approx 8.33$$

θ	0	23.57°	30°
$\frac{dT}{d\theta}$	-25	0	8.33
T		Min	

(iv)

The minimum time occurs when $\sin \theta = 0.4$ and hence $\cos \theta = \sqrt{0.84}$

This is about 100.6 seconds.

Also Accept 1 min 36 seconds

(iii)

1 for $\frac{dT}{d\theta}$

1 for getting
 $\sin \theta = 0.4$

1 for θ

1 for testing

(iv)

1 for correct
answer